

# MoPA: Mobile Manipulation via Subsystem-Specific Perception Alignment and Action Coordination

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**Abstract**—Mobile manipulation requires perceptual evidence at different spatial scales for base motion and arm control, while their actions remain kinematically coupled. Existing policies often specialize action generation but condition heterogeneous action branches on a shared perceptual representation, leaving subsystem-level perception–action correspondence implicit. We present MoPA, a framework that aligns perceptual conditioning with mobility and manipulation while retaining action-level coordination. Dual Perceptual Streams use two mutually masked query banks to derive separate perceptual representations from a shared vision–language context. Perception2Action Adaptation jointly updates each query bank and its matched action stream at every layer of a structured Mixture-of-Transformers decoder, while allowing the two action streams to exchange information. Coupled conditional flow matching learns a joint vector field for synchronous generation of both action chunks. On the ManiSkill-HAB benchmark, MoPA achieves state-of-the-art performance across all three task suites. Across four real-world tasks, MoPA achieves a mean full-task success rate of 76.3%, outperforming the strongest evaluated baseline  $\pi_{0.5}$  by 12.5 percentage points. Ablation studies and analyses further support the effectiveness of the proposed design.

## I. INTRODUCTION

Robots deployed in unstructured household environments must move between workspaces and interact with objects as their egocentric views continuously change. Unlike fixed-base tabletop manipulation, mobile manipulation requires joint control of a mobile base and a manipulator. The base relies on scene-level geometry for repositioning, whereas the manipulator requires local geometry and interaction state for precise end-effector control. Their action spaces differ, yet their motions are coupled through a single kinematic chain: base-pose errors propagate to the end effector, while arm reachability constrains feasible base motion. A policy must therefore organize perceptual evidence by subsystem and generate coordinated whole-body actions [1], [2], [3].

Recent work improves perception under changing viewpoints by combining multiview observations with geometric, temporal, and semantic context [4], [5], [6], [7], [8]. Complementary work structures action generation for heterogeneous control through hierarchical or parallel decoders, token routing, and specialized Transformer streams [9], [10], [11]. Together, these directions expand the evidence available to a policy and tailor action computation to heterogeneous control, yet the action branches generally remain conditioned on a shared or globally fused representation.

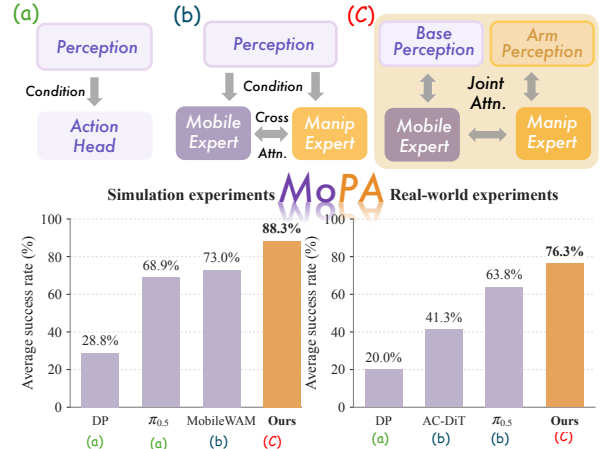


Fig. 1. Mobile-manipulation paradigms and performance. (a) Shared perception conditions a unified action head. (b) Separate base and manipulation experts share perceptual conditioning. (c) MoPA combines subsystem-specific perception with action-level coordination. Bottom: average success rates in simulation and real-world experiments.

Despite these advances, subsystem-level perception–action alignment remains unresolved. Mobile manipulation requires selecting perceptual evidence according to each subsystem’s control objective and generating actions through subsystem-specific pathways while preserving coordination. Existing methods typically address these requirements separately, linking perception and action through a global representation. During simultaneous base and arm motion, that representation must encode both scene layout and local interaction geometry, despite the different spatial scales emphasized by the two subsystems. Even with separate action heads, losses from both branches can update shared projection or fusion layers, potentially inducing gradient conflicts in multi-task optimization [12]. Separating output parameters therefore does not guarantee that the action experts receive subsystem-specific perceptual conditioning.

Existing perception–action interfaces also struggle to support this separation. Directly supplying fused features to multiple action branches gives all experts the same, undifferentiated conditioning. Allowing action tokens to retrieve information independently from the full perceptual sequence requires repeated access to a large token set and provides no persistent intermediate representation aligned with a specific action space. We therefore argue that mobile manipulation requires a mechanism that bridges shared perception and

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heterogeneous action experts. Such a mechanism should derive subsystem-specific representations from a common vision–language context, update each representation as its corresponding actions are generated, and retain coordination between the base and manipulator at the action level.

As summarized in Fig. 1, we introduce MoPA, a subsystem-aligned mobile-manipulation framework with two complementary modules. *Dual Perceptual Streams* connect a shared Qwen3-VL context [13] to heterogeneous action experts. Given the same head- and wrist-camera RGB images and language instruction, isolated *Manip Query* and *Mobile Query* banks aggregate subsystem-relevant evidence. A structured mask prevents direct interaction between the banks, yielding distinct perceptual representations before action decoding.

*Perception2Action Adaptation* adapts these perceptual representations to heterogeneous action generation while coordinating whole-body control. A structured Mixture-of-Transformers decoder assigns separate action tokens and parameter sets to manipulation and mobility. Each query bank interacts bidirectionally only with its matched action stream, while the two action streams exchange their evolving control states. Layerwise query–action updates transform perceptual summaries into action-conditioned states. We use coupled flow matching to train the model and jointly sample the two action chunks. Dual Perceptual Streams thus provide separate perceptual conditioning for the two subsystems, while Perception2Action Adaptation adapts and coordinates their action generation.

On ManiSkill-HAB [14], MoPA achieves the highest macro-averaged skill success rates among the compared methods on SetTable (**88.3%**), TidyHouse (**72.2%**), and PrepareGroceries (**67.8%**). These results exceed the strongest baseline on each suite by 4.5, 17.0, and 7.1 percentage points, respectively.

Our contributions are:

- We characterize the lack of explicit subsystem-level perception–action correspondence and explain why separating action heads alone is insufficient.
- We propose *Dual Perceptual Streams*, which use isolated *Manip Query* and *Mobile Query* banks to derive subsystem-specific perceptual representations from a shared vision–language context.
- We propose *Perception2Action Adaptation*, which jointly updates these representations with their matched action tokens and coordinates separate action streams.
- We achieve state-of-the-art performance on all three ManiSkill-HAB task suites and validate our design through extensive ablation studies and exploratory experiments.

## II. RELATED WORK

**Visuomotor Policies.** For mobile manipulation, visuomotor policies directly map onboard observations and robot state to whole-body commands. Mobile ALOHA and HarmonicMM jointly model mobile manipulation, while SPIN

adds active camera motion [1], [2], [3]. CausalMoMa introduces causal credit assignment, whereas VBC and UMI on Legs decompose whole-body control into body/base targets and task-frame end-effector trajectories [15], [16], [17]. Representative perception designs such as DSPv2, AC-DiT, and InCoM align or reweight multiview semantic–geometric features; SBP maintains a persistent 3D map, and GeoHAT routes reliable visual tokens to base and arm actions [4], [5], [6], [7], [8]. Mobile UMI further factorizes manipulation and base trajectories [9]. These methods improve grounding and execution, but typically select subsystem-specific evidence within the action decoder rather than maintaining it as persistent conditioning representations before action generation.

**Vision–Language–Action Policies.** VLA policies incorporate pretrained vision–language priors into instruction-conditioned control.  $\pi_{0.5}$  scales training on heterogeneous data to support long-horizon tasks. MoManipVLA transfers fixed-base policies by optimizing base waypoints and end-effector trajectories, whereas Mobi- $\pi$  searches at deployment for an in-distribution base pose for an existing visuomotor policy [18], [19], [20]. Large-scale systems such as G0 and G0.5 couple planning, reasoning, and action tokenization [21], [22]. SG-VLA, EchoVLA, SERF, and PanoVLA enrich spatial or temporal context with multiview observations, memory, online maps, or panoramic views; AnchorVLA and MolmoBOT target efficient closed-loop generation and sim-to-real transfer, respectively [23], [24], [25], [26], [27], [28]. Nevertheless, whole-body policies generally use a shared action interface, leaving the correspondence between perceptual evidence and base/manipulation action spaces implicit.

**World–Action Models.** WAMs augment current observations with predictions of future visual or latent dynamics. DreamZero jointly predicts future video and actions, MotionWAM uses video-model denoising features for control, and  $\omega$ -0 conditions diffusion actions on future-observation embeddings [29], [30], [31]. For mobile manipulation, MobileWAM routes computation among shared, mobility, and manipulation experts, ABot-M0.5 combines latent actions with a Mixture-of-Transformers, and DECOWAM separates base and arm latents [10], [11], [32]. World–Ego Modeling and DreamTrajectory further factorize world/ego prediction and rank candidate whole-body trajectories, respectively [33], [34]. These methods specialize mainly through future latents or action experts; MoPA instead provides an explicit, persistent interface between shared perceptual context and coordinated action streams.

## III. METHOD

This section presents MoPA, a subsystem-aligned perception–action framework for mobile manipulation. We first define the observation and action spaces (Sec. III-A) and outline the architecture (Sec. III-B). We then introduce Dual Perceptual Streams, which derive branch-specific perceptual representations from shared vision–language tokens (Sec. III-C), and Perception2Action Adaptation, which models and coordinates the two action streams (Sec. III-D). Finally, we

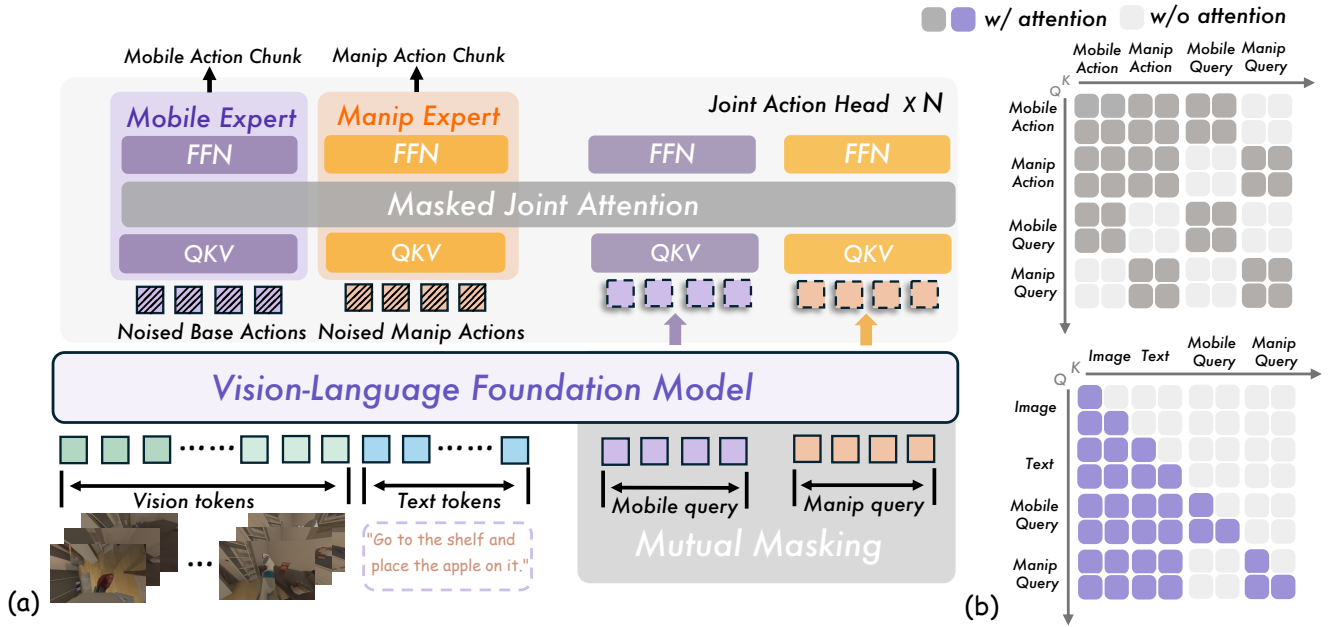


Fig. 2. Overview of MoPA. (a) A shared vision–language foundation model produces separate *Mobile Query* and *Manip Query* streams that condition dedicated Base and Manipulation experts in a joint action head. (b) Structured visibility patterns in the VLM and action head: direct attention between the query streams is masked, each query stream connects to its matched action stream, and the two action streams remain coupled for whole-body coordination.

describe the coupled conditional flow-matching objective and synchronous inference procedure (Sec. III-E).

#### A. Problem Formulation

At time  $t$ , the policy receives head- and wrist-camera RGB images  $I_t^{\text{head}}$  and  $I_t^{\text{wrist}}$ , a proprioceptive state  $\mathbf{s}_t \in \mathbb{R}^{d_s}$ , and a language instruction  $\ell$ . Following the input/output convention of AC-DiT [5], it predicts a whole-body action chunk of horizon  $H$ ,  $\mathbf{A}_t = (\mathbf{A}_t^{\text{man}}, \mathbf{A}_t^{\text{mob}})$ . For branch  $k \in \{\text{man}, \text{mob}\}$ ,  $\mathbf{A}_t^k = (\mathbf{a}_t^k, \dots, \mathbf{a}_{t+H-1}^k) \in \mathbb{R}^{H \times d_k}$ . The policy models the conditional distribution

$$p_{\Theta}(\mathbf{A}_t^{\text{man}}, \mathbf{A}_t^{\text{mob}} \mid I_t^{\text{head}}, I_t^{\text{wrist}}, \mathbf{s}_t, \ell). \quad (1)$$

The manipulation and mobility branches control the arm/gripper and torso/base, respectively. Joint modeling allows the policy to learn their coordination directly from whole-body demonstrations.

#### B. Method Overview

MoPA has three stages. First, Qwen3-VL encodes the head-view image, wrist-view image, and language instruction into a shared vision–language token sequence. Two isolated query banks, termed the *Manip Query* and *Mobile Query*, extract branch-specific perceptual streams from this sequence. The proprioceptive state bypasses the VLM and is encoded separately in the two action branches.

Next, Perception2Action Adaptation uses a structured Mixture-of-Transformers (MoT) decoder to pair each perceptual stream with its corresponding action stream while allowing the two action streams to exchange information. Finally, coupled conditional flow matching learns a joint vector field

for synchronous generation of both action chunks. Dual Perceptual Streams thus determine the perceptual information available to each subsystem, while Perception2Action Adaptation models their actions and coordination. The complete architecture and its information flow are shown in Fig. 2.

#### C. Dual Perceptual Streams

The shared vision–language context contains scene-level cues for mobility and local interaction cues for manipulation. When both action experts receive the same representation, each must extract the information relevant to its control objective from an undifferentiated conditioning signal. Dual Perceptual Streams construct a dedicated perceptual representation for each subsystem before action decoding.

Let  $\mathbf{P}_t$  denote the shared vision–language token sequence, and let  $\mathbf{Q}^{\text{Manip}}$  and  $\mathbf{Q}^{\text{Mob}}$  be the learned Manip Query and Mobile Query banks. We append both banks to  $\mathbf{P}_t$ :

$$\mathbf{Z}_t = [\mathbf{P}_t, \mathbf{Q}^{\text{Manip}}, \mathbf{Q}^{\text{Mob}}]. \quad (2)$$

Let  $\mathbf{H}_t^{\text{Manip}}$  and  $\mathbf{H}_t^{\text{Mob}}$  denote the final VLM hidden states at the positions of the two query banks. As shown in the lower matrix of Fig. 2(b), both banks attend to  $\mathbf{P}_t$ , while direct attention between them is masked. They therefore aggregate the same context through separate paths. Distinct learned embeddings, separate aggregation paths, and supervision from the corresponding action branches encourage each bank to retain information relevant to its control objective. The resulting states are routed only to their matched action streams, providing distinct perceptual representations before action prediction. Perception2Action Adaptation then handles cross-subsystem coordination.

#### D. Perception2Action Adaptation

Subsystem-specific perceptual representations alone do not ensure coordinated whole-body control: independently decoded manipulation and mobility commands can still conflict. Perception2Action Adaptation addresses this issue with a structured MoT decoder that uses separate parameters for the two action branches while coupling the evolving states of both action streams.

Separate projectors first map the perceptual states into the feature space of the action model:

$$\overline{\mathbf{Q}}_{t,0}^q = \Phi^q(\mathbf{H}_t^q), \quad q \in \{\text{Manip}, \text{Mob}\}. \quad (3)$$

For each branch  $k \in \{\text{man}, \text{mob}\}$ , we concatenate the proprioceptive and action tokens at flow time  $\tau$  to form the initial branch sequence:

$$\mathbf{X}_{\tau,0}^k = [E_s^k(\mathbf{s}_t) + \mathbf{e}_s^k, E_a^k(\mathbf{A}_\tau^k, \tau) + \mathbf{E}_{\text{pos}}^k]. \quad (4)$$

The projectors  $\Phi^q$  and encoders  $E_s^k, E_a^k$  are branch-specific;  $\mathbf{e}_s^k$  and  $\mathbf{E}_{\text{pos}}^k$  are the state-token type embedding and action-token positional embeddings, respectively. Here,  $q = \text{Manip}$  and  $q = \text{Mob}$  index the query streams, while  $k = \text{man}$  and  $k = \text{mob}$  index their matched action branches. The continuous flow time is represented by a timestep embedding in the action encoder and in each decoder block.

The joint action head processes the two query streams and two action streams using the structured connectivity in the upper matrix of Fig. 2(b). Each query bank interacts with its corresponding action stream, while the two action streams exchange information to coordinate whole-body control.

Let  $L$  denote the number of decoder blocks. For  $l = 0, \dots, L-1$ , the  $l$ -th block, represented by  $\mathcal{B}_l$ , performs the joint update

$$\begin{aligned} & (\overline{\mathbf{Q}}_{t,l+1}^{\text{Manip}}, \overline{\mathbf{Q}}_{t,l+1}^{\text{Mob}}, \mathbf{X}_{\tau,l+1}^{\text{man}}, \mathbf{X}_{\tau,l+1}^{\text{mob}}) \\ &= \mathcal{B}_l(\overline{\mathbf{Q}}_{t,l}^{\text{Manip}}, \overline{\mathbf{Q}}_{t,l}^{\text{Mob}}, \mathbf{X}_{\tau,l}^{\text{man}}, \mathbf{X}_{\tau,l}^{\text{mob}}; \tau). \end{aligned} \quad (5)$$

Each stream has its own projection, normalization, and feed-forward parameters. At each velocity-field evaluation, the query states are initialized using Eq. (3) and updated jointly with the action tokens at every layer. They thus evolve from perceptual summaries into action-conditioned control states within that evaluation. The final action tokens are decoded into flow velocities for manipulation and mobility.

#### E. Coupled Conditional Flow Matching

We train the coupled decoder to model a conditional vector field over the joint whole-body action space. For each branch  $k$ , let  $\mathbf{A}_1^k \equiv \mathbf{A}_t^k$  denote a clean demonstration chunk and sample an independent noise chunk  $\mathbf{A}_0^k \in \mathbb{R}^{H \times d_k}$  with i.i.d. standard Gaussian entries. At a flow time  $\tau \sim \mathcal{U}(0, 1)$  shared by both branches, the linear interpolation and its constant target velocity are

$$\mathbf{A}_\tau^k = (1 - \tau)\mathbf{A}_0^k + \tau\mathbf{A}_1^k, \quad \mathbf{U}^k = \mathbf{A}_1^k - \mathbf{A}_0^k. \quad (6)$$

Conditioned on the projected query states and encoded branch sequences, the decoder jointly predicts a velocity

$\hat{\mathbf{U}}^k \in \mathbb{R}^{H \times d_k}$  for each branch. Because the action streams exchange information in every decoder block, each prediction can depend on the current state of the other branch. We optimize the joint vector field with a branch-normalized flow-matching objective:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{A}_1, \mathbf{A}_0, \tau} \left[ \sum_{k \in \{\text{man}, \text{mob}\}} \frac{\lambda_k}{H d_k} \left\| \hat{\mathbf{U}}^k - \mathbf{U}^k \right\|_F^2 \right], \quad (7)$$

where  $\lambda_k$  balances the branch contributions, and the squared Frobenius norm sums the squared errors over the  $H \times d_k$  entries. Normalization by  $H d_k$  prevents the higher-dimensional branch from dominating the objective solely because it has more action channels. The vision-language backbone, Dual Perceptual Streams, and Perception2Action Adaptation are optimized end to end with Eq. (7).

At inference, both action chunks start from independent Gaussian noise and are integrated synchronously using  $N$  Euler steps of size  $1/N$ . At each shared flow time  $\tau_n = n/N$ , the decoder predicts both velocities from the current chunks, then updates both chunks simultaneously by adding their predicted velocities scaled by  $1/N$ . We concatenate the final manipulation and mobility chunks along the action dimension to form the whole-body chunk. The robot executes its initial actions and replans from updated observations.

## IV. EXPERIMENTS

We evaluate MoPA on simulation benchmarks (Sec. IV-A), examine its core components through ablations (Sec. IV-B), explore query capacity and representational differentiation (Sec. IV-C), and assess real-world performance (Sec. IV-D).

#### A. Simulation Benchmark

**Experimental Setup.** We evaluate MoPA on ManiSkill-HAB [14], a household mobile-manipulation benchmark built on ManiSkill3 and SAPIEN. It combines ReplicaCAD apartment scenes with YCB object models and supports whole-body control of a Fetch mobile manipulator. Our evaluation covers three suites: SetTable, TidyHouse, and Prepare-Groceries. For SetTable, we evaluate seven skills: picking and placing an apple and a bowl, opening a fridge, and opening and closing a kitchen drawer. TidyHouse involves object rearrangement across open receptacles, whereas PrepareGroceries involves transferring objects between a fridge and kitchen counters. Each of these two suites contains 18 pick/place skills across nine object categories. Together, the suites require coordinated base and arm motion in cluttered scenes for both object rearrangement and interaction with articulated objects.

**Implementation Details.** We train one policy per suite using 1,000 successful demonstrations per skill. The inputs comprise RGB images from the head and wrist cameras, proprioception, and language instructions. We use an action chunk length of  $H = 4$ , with an 11-D whole-body action at each time step. We train for 100k optimization steps using AdamW, cosine learning-rate decay, and a batch size of 64.

TABLE I

SETTABLE BENCHMARK RESULTS. ENTRIES REPORT MEAN SUCCESS RATES (%), WITH STANDARD DEVIATIONS IN GRAY. FOR EACH METRIC, RANKINGS AMONG EVALUATED METHODS ARE MARKED AS **BEST** AND **SECOND BEST**. \* DENOTES DEPTH INPUT.

| Method             | Pick Apple                       | Pick Bowl                        | Place Apple                      | Place Bowl                       | Open Fridge                      | Open Drawer                      | Close Drawer                      | Mean        |
|--------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|-----------------------------------|-------------|
| DP3* [35]          | 0.0 $\pm$ 0.0                    | 20.0 $\pm$ 2.4                   | 31.0 $\pm$ 0.8                   | 32.0 $\pm$ 0.8                   | 0.0 $\pm$ 0.0                    | 0.0 $\pm$ 0.0                    | 68.0 $\pm$ 0.0                    | 21.6        |
| ACT [36]           | 28.0 $\pm$ 2.2                   | 28.0 $\pm$ 2.4                   | 8.7 $\pm$ 3.3                    | 13.0 $\pm$ 0.8                   | 2.0 $\pm$ 2.2                    | 0.0 $\pm$ 0.0                    | 85.7 $\pm$ 1.2                    | 23.6        |
| DP [37]            | 21.3 $\pm$ 3.3                   | 20.7 $\pm$ 3.3                   | 28.0 $\pm$ 8.0                   | 69.3 $\pm$ 3.3                   | 7.3 $\pm$ 5.8                    | 0.0 $\pm$ 0.0                    | 55.0 $\pm$ 5.7                    | 28.8        |
| RDT-1B [38]        | 12.0 $\pm$ 11.3                  | 10.7 $\pm$ 6.8                   | 32.0 $\pm$ 5.7                   | 18.7 $\pm$ 5.0                   | 82.7 $\pm$ 10.5                  | 44.0 $\pm$ 8.6                   | <b>100.0 <math>\pm</math> 0.0</b> | 42.9        |
| AC-DiT* [5]        | 33.3 $\pm$ 1.9                   | 36.0 $\pm$ 6.5                   | 33.3 $\pm$ 9.4                   | 17.3 $\pm$ 6.8                   | 90.7 $\pm$ 5.0                   | 81.3 $\pm$ 6.8                   | <b>97.3 <math>\pm</math> 1.9</b>  | 55.6        |
| $\pi_0$ [39]       | 26.6 $\pm$ 3.8                   | 26.6 $\pm$ 4.5                   | 48.9 $\pm$ 5.2                   | 56.8 $\pm$ 5.9                   | 90.5 $\pm$ 1.9                   | 75.5 $\pm$ 2.5                   | 88.7 $\pm$ 2.0                    | 59.1        |
| AnchorVLA [27]     | 22.7 $\pm$ 0.9                   | 44.5 $\pm$ 0.8                   | 64.3 $\pm$ 0.8                   | 63.8 $\pm$ 2.0                   | 88.9 $\pm$ 0.8                   | —                                | <b>100.0 <math>\pm</math> 0.0</b> | 64.0        |
| $\pi_{0.5}$ [18]   | 44.2 $\pm$ 2.3                   | 45.3 $\pm$ 1.9                   | 57.1 $\pm$ 2.9                   | 65.7 $\pm$ 1.3                   | 92.9 $\pm$ 0.8                   | 86.6 $\pm$ 3.4                   | 90.5 $\pm$ 1.5                    | 68.9        |
| MobileWAM [10]     | 46.0 $\pm$ 0.8                   | 46.0 $\pm$ 2.6                   | 63.7 $\pm$ 3.2                   | 64.7 $\pm$ 1.2                   | <b>99.3 <math>\pm</math> 0.5</b> | <b>91.0 <math>\pm</math> 0.8</b> | <b>100.0 <math>\pm</math> 0.0</b> | 73.0        |
| GeoHAT* [8]        | <b>82.3 <math>\pm</math> 2.5</b> | 69.3 $\pm$ 1.3                   | 60.0 $\pm$ 1.6                   | 78.0 $\pm$ 0.0                   | 83.7 $\pm$ 0.5                   | 86.0 $\pm$ 0.0                   | 95.3 $\pm$ 0.5                    | 79.3        |
| InCoM* [6]         | 59.4 $\pm$ 4.3                   | <b>84.1 <math>\pm</math> 4.6</b> | <b>84.1 <math>\pm</math> 5.4</b> | <b>82.5 <math>\pm</math> 5.6</b> | 87.3 $\pm$ 2.8                   | 88.9 $\pm$ 2.3                   | <b>100.0 <math>\pm</math> 0.0</b> | <b>83.8</b> |
| <b>MoPA (Ours)</b> | <b>84.0 <math>\pm</math> 0.6</b> | <b>89.0 <math>\pm</math> 2.0</b> | <b>73.0 <math>\pm</math> 0.6</b> | <b>84.0 <math>\pm</math> 2.5</b> | <b>98.0 <math>\pm</math> 0.6</b> | <b>94.0 <math>\pm</math> 1.0</b> | 96.0 $\pm$ 0.6                    | <b>88.3</b> |

TABLE II

MEAN SUCCESS RATES (%) ON TIDYHOUSE AND PREPAREGROCERIES. STANDARD DEVIATIONS ARE GRAY; **BEST** AND **SECOND BEST** ARE MARKED PER COLUMN.

| Method             | TidyHouse                        |                                  |             | PrepareGroceries                 |                                  |             |
|--------------------|----------------------------------|----------------------------------|-------------|----------------------------------|----------------------------------|-------------|
|                    | Pick                             | Place                            | Mean        | Pick                             | Place                            | Mean        |
| ACT [36]           | 2.2 $\pm$ 0.8                    | 31.6 $\pm$ 3.2                   | 16.9        | 2.0 $\pm$ 1.1                    | 27.5 $\pm$ 2.9                   | 14.8        |
| DP [37]            | 0.0 $\pm$ 0.0                    | 30.3 $\pm$ 3.0                   | 15.2        | 0.4 $\pm$ 0.3                    | 17.7 $\pm$ 1.9                   | 9.1         |
| DP3 [35]           | 0.0 $\pm$ 0.0                    | 61.0 $\pm$ 0.0                   | 30.5        | 0.0 $\pm$ 0.0                    | 31.3 $\pm$ 1.2                   | 15.7        |
| DSPv2 [4]          | 1.3 $\pm$ 0.3                    | 42.1 $\pm$ 3.1                   | 21.7        | 0.9 $\pm$ 0.6                    | 32.8 $\pm$ 2.9                   | 16.9        |
| InCoM [6]          | 16.7 $\pm$ 1.9                   | <b>78.9 <math>\pm</math> 2.9</b> | 47.8        | 15.0 $\pm$ 2.4                   | <b>65.9 <math>\pm</math> 3.1</b> | 40.5        |
| GeoHAT [8]         | 30.3 $\pm$ 0.5                   | <b>73.3 <math>\pm</math> 1.7</b> | 51.8        | 19.7 $\pm$ 4.6                   | 60.7 $\pm$ 2.5                   | 40.2        |
| $\pi_{0.5}$ [18]   | <b>51.4 <math>\pm</math> 1.3</b> | 58.9 $\pm$ 2.4                   | <b>55.2</b> | <b>59.3 <math>\pm</math> 2.1</b> | 62.1 $\pm$ 3.5                   | <b>60.7</b> |
| <b>MoPA (Ours)</b> | <b>73.9 <math>\pm</math> 0.8</b> | 70.4 $\pm$ 2.3                   | <b>72.2</b> | <b>66.6 <math>\pm</math> 1.2</b> | <b>69.0 <math>\pm</math> 1.6</b> | <b>67.8</b> |

At inference, each action chunk is sampled using four Euler integration steps. We execute the first two actions at 20 Hz before replanning from updated observations.

We evaluate each skill in three runs, each comprising 100 held-out episodes with a maximum of 200 steps per episode. We report the mean and standard deviation of success rates across the three runs. Each suite score is the macro-average of its skill success rates.

**Baselines.** We compare representative end-to-end imitation-learning methods evaluated on ManiSkill-HAB across three paradigms: *Visuomotor Policies*, including DP3 [35], ACT [36], DP [37], DSPv2 [4], AC-DiT [5], GeoHAT [8], InCoM [6], and RDT-1B [38]; *Vision–Language–Action Policies*, including  $\pi_0$  [39],  $\pi_{0.5}$  [18], and AnchorVLA [27]; and *World–Action Models*, represented by MobileWAM [10]. Methods that use depth information as input are marked with \* in Table I.

**Results and Analysis.** MoPA achieves the highest macro-averaged skill success rate among the compared methods on all three suites. On SetTable, it reaches 88.3%, exceeding InCoM by 4.5 percentage points and leading on four of seven skills (Table I). On TidyHouse and PrepareGroceries, it achieves 72.2% and 67.8%, respectively, exceeding the strongest baseline  $\pi_{0.5}$  (55.2% and 60.7%) by 17.0 and 7.1 percentage points (Table II).

Gains over  $\pi_{0.5}$  span picking and placement on both suites,

with a larger TidyHouse gain on Pick than Place (22.5 vs. 11.5 percentage points). InCoM still leads on SetTable’s Place Apple and TidyHouse’s Place category.

Related baselines incorporate stage-adaptive perception [6], geometry-aware visual selection [8], or specialized action experts [10]. On SetTable, MobileWAM reaches 99.3% on Open Fridge and 100.0% on Close Drawer but 46.0% on both picking tasks, where MoPA achieves 84.0% on Pick Apple and 89.0% on Pick Bowl. This gap motivates aligning perceptual conditioning with each subsystem alongside action specialization.

MoPA constructs separate perceptual streams and refines them during coordinated action generation. In the example in Fig. 4, Mobile Query emphasizes the floor and surrounding scene structure, whereas Manip Query focuses on the shelf around the target object, consistent with their intended roles. The performance drops for Shared Query, w/o Joint Attention, and w/o Corresponding Query Access (Table III) support subsystem-specific conditioning and its adaptation during action decoding.

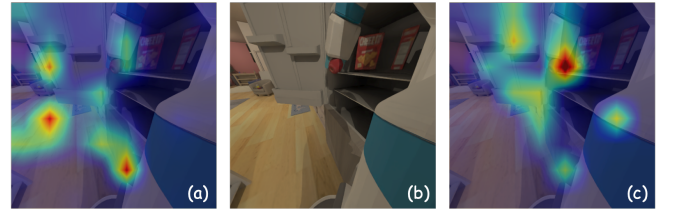


Fig. 4. Query attention visualization. (a) Mobile Query attention, (b) the RGB observation, and (c) Manip Query attention.

### B. Ablation Studies

We evaluate Dual Perceptual Streams and Perception2Action Adaptation on SetTable (Table III). We first examine how perceptual representations are constructed, then how they are updated and matched to action branches. All variants use the same backbone and training and evaluation protocols, and retain both action experts and interaction between their streams.

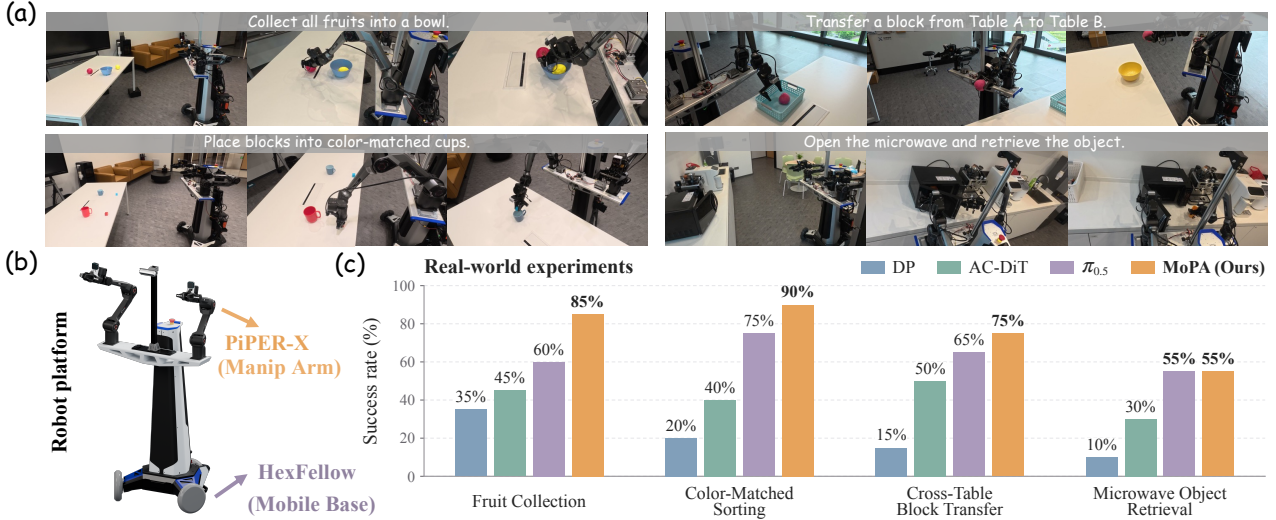


Fig. 3. Real-world evaluation of MoPA. (a) Execution sequences for fruit collection, color-matched sorting, cross-table block transfer, and microwave object retrieval. (b) Robot platform. (c) Success rates compared with DP, AC-DiT, and  $\pi_{0.5}$ .

a) *Perceptual conditioning*: We first compare direct VLM conditioning with learned query aggregation. Direct VLM Conditioning replaces queries with VLM hidden states while retaining branch-specific projections and joint condition-action updates. Shared Query supplies one learned query bank to both branches, increasing the success rate from 58.4% to 74.7%. Separating the queries by subsystem further raises the success rate to 88.3% without changing the total learned query capacity. These comparisons support both learnable extraction of perceptual information and subsystem-specific conditioning; query aggregation alone does not explain the overall improvement.

b) *Joint query-action adaptation*: With subsystem-specific representations in place, we next examine whether they benefit from action-dependent updates. The w/o Joint Attention variant keeps query parameters trainable and preserves matched query access, but fixes query hidden states within each action-head evaluation. The success rate drops to 70.3%, 18.0 percentage points below that of the full model. Since both action experts and their interaction are retained, this result supports adapting perceptual representations to evolving actions rather than supplying static summaries.

c) *Subsystem-aligned access*: We next examine which query states each action stream should access during joint updating. The w/o Corresponding Query Access variant retains both query banks and joint updates but allows each action stream to read both banks. Unlike Shared Query, it changes action-side access while preserving separate query aggregation. The success rate drops to 66.3%, 22.0 percentage points below that of the full model. This result supports maintaining explicit query-action correspondence during decoding, beyond constructing separate query banks and updating them jointly with actions.

### C. Exploratory Experiments

We investigate two research questions (RQs):

TABLE III  
COMPONENT ABLATIONS AND QUERY BANK SIZE ANALYSIS ON SETTABLE.

| Variant                        | Avg. SR     | $\Delta$ SR |
|--------------------------------|-------------|-------------|
| Direct VLM Conditioning        | 58.4        | ↓ 29.9      |
| Shared Query                   | 74.7        | ↓ 13.6      |
| w/o Joint Attention            | 70.3        | ↓ 18.0      |
| w/o Corresponding Query Access | 66.3        | ↓ 22.0      |
| MoPA w/ Queries per bank = 4   | 74.6        | ↓ 13.7      |
| MoPA w/ Queries per bank = 16  | 82.5        | ↓ 5.8       |
| MoPA w/ Queries per bank = 8   | <b>88.3</b> | –           |

- **RQ1 (Query Bank Size)**: Does using eight queries per action branch yield the highest success rate among the tested configurations?
- **RQ2 (Query Differentiation)**: Do the Mobile Query and Manip Query banks exhibit distinct representations during inference?

a) *RQ1: Effect of Query Bank Size*: We vary the number of queries per bank,  $q \in \{4, 8, 16\}$ , corresponding to 8, 16, and 32 total queries, while keeping the remaining architecture and training settings fixed. The final three rows of Table III report average success rates on SetTable.

Average success increases from 74.6% at  $q = 4$  to 88.3% at  $q = 8$ , then decreases to 82.5% at  $q = 16$ . We therefore adopt the eight-query configuration, which achieves the highest success rate among the tested settings. This sweep assesses sensitivity to query capacity, whereas Shared Query tests the value of subsystem differentiation with a fixed learned query budget.

b) *RQ2: Query Representation Differentiation*: To investigate the representational differences between Manip Queries and Mobile Queries, we collect query states during

inference on a 20% subset of the dataset and visualize them using a cosine-similarity heatmap and t-SNE (Fig. 5). The heatmap shows low or negative cosine similarity for many cross-type query pairs, indicating differences in their feature directions. Consistently, the t-SNE projection shows that Manip Query and Mobile Query states predominantly occupy distinct regions, with partial overlap. Together, these observations provide qualitative evidence of differentiated representations between the two query types during inference, consistent with their intended roles in manipulation and mobility.

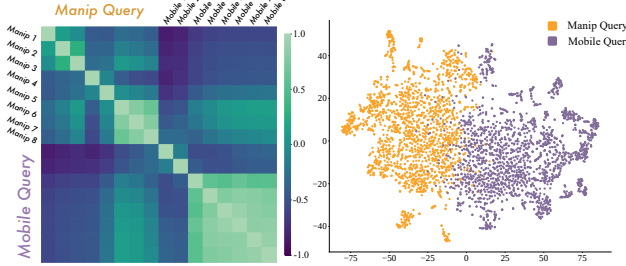


Fig. 5. Query representations during inference. Left: cosine-similarity heatmap. Right: t-SNE projection, with orange and purple denoting Manip Query and Mobile Query states, respectively.

#### D. Real-World Experiments

**Experimental Setup.** Our platform (Fig. 3(b)) comprises a HexFellow Trigger-A3 omnidirectional mobile base, an IOTA vertical lift, and two AgileX PiPER-X six-DoF arms. The platform has three RGB-D cameras: one on the head and one on each wrist.

We evaluate four tasks, with representative execution sequences shown in Fig. 3(a):

- *Fruit Collection:* The robot approaches a table and places all fruit into a bowl.
- *Color-Matched Sorting:* The robot approaches a table and places blocks into color-matched cups.
- *Cross-Table Block Transfer:* The robot picks up a block at Table A, navigates to Table B, and places it there.
- *Microwave Object Retrieval:* The robot approaches a microwave, opens its door, and retrieves an object.

These tasks require repeated pick-and-place, color-based object–receptacle matching, and multi-stage coordination across workspaces or during articulated-object interaction. We collect 200 demonstrations per task.

**Implementation Details.** We retain the action chunk length of  $H = 4$  used in simulation. The whole-body action at each time step is 17-dimensional and includes three base velocity components  $[v_x, v_y, \omega_z]$  for planar translation and yaw rotation. We compare MoPA against Diffusion Policy (DP) [37], AC-DiT [5], and  $\pi_{0.5}$  [18] on the four real-world tasks. MoPA, AC-DiT, and  $\pi_{0.5}$  each use one policy jointly trained on all four tasks for 50k optimization steps with a batch size of 64. For DP, we train a separate policy per task for 100 epochs with a batch size of 128. Each method

is evaluated over 20 trials per task, and we report full-task success rates.

**Results and Analysis.** Figure 3(c) reports task success rates. MoPA achieves success rates of 85.0%, 90.0%, 75.0%, and 55.0% on Fruit Collection, Color-Matched Sorting, Cross-Table Block Transfer, and Microwave Object Retrieval, respectively. Its average success rate of 76.3% exceeds that of the strongest baseline,  $\pi_{0.5}$ , by 12.5 percentage points. MoPA achieves the highest success rate on the first three tasks and ties with  $\pi_{0.5}$  on microwave retrieval.

Fruit Collection and Color-Matched Sorting require repeated object selection and precise placement, whereas Cross-Table Block Transfer combines grasping and placement with movement between workspaces. Dual Perceptual Streams provides separate perceptual conditions for mobility and manipulation, and Perception2Action refines these conditions alongside action generation while coordinating the two action streams. The high success rates on these tasks demonstrate that MoPA can combine fine-grained interaction and cross-workspace execution within a single multi-task policy.

#### V. CONCLUSION

We presented MoPA, a mobile-manipulation framework that connects shared vision–language representations to coordinated action generation through subsystem-aligned perceptual streams. Dual Perceptual Streams construct distinct perceptual representations for mobility and manipulation, while Perception2Action refines them jointly with their corresponding action streams and preserves communication between the action streams. Experiments on three ManiSkill-HAB suites and four real-world tasks show higher average success rates than the evaluated baselines. Ablation studies and analyses support the complementary roles of separate perceptual conditioning, action-dependent query updates, and matched query access. These findings support explicit perception–action correspondence as a useful design principle for coordinated mobile manipulation. Performance remains task-dependent, as illustrated by the tie with the strongest evaluated baseline on microwave retrieval, motivating further evaluation across more diverse environments and interaction demands.

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